

Preliminary Exploration of the Effects of Ceiling Height and Spatial Sharing Methods on Cognitive Load in Collaborative Virtual Reality

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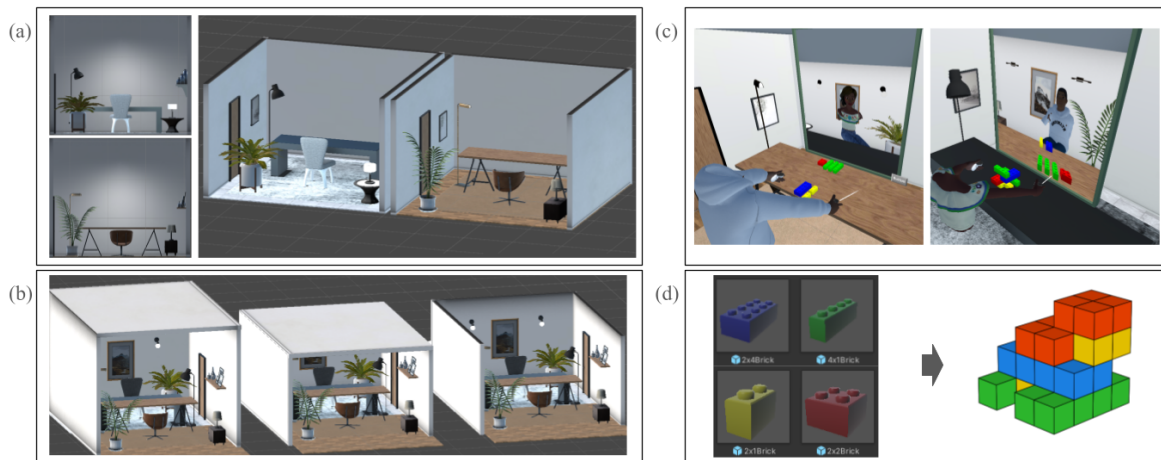


Figure 1: Visualization of collaborative shared environments and task components. (a) Example of the two virtual reality environments used in the experiment. Each space maintains comparable scale and functionality. (b) Cross-sectional views of a shared space using an overlay, illustrating scenarios from left to right with a high-ceiling, a low-ceiling, and an open-ceiling environment. (c) Example of a shared space with low-ceiling using a portal, illustrating the executor's view on the left and the instructor's view on the right. (d) The left side presents the four types of blocks utilized in the collaborative construction task, while the right side shows an example of the desired block configuration that participants are required to assemble.

ABSTRACT

Virtual Reality (VR) environments increasingly support shared spaces in which multiple users interact across physically separate locations. While prior studies have explored usability in shared environments, the impact of spatial design factors on users' cognitive load remains insufficiently understood. This study investigates how ceiling height and spatial sharing methods influence cognitive load during interactions in VR shared spaces. We conducted a controlled user study examining combinations of three ceiling height conditions (low, high, and open) and two spatial sharing methods (portal and overlay). The results indicate that ceiling height significantly affects overall cognitive load, with high- and open- ceiling conditions yielding lower load compared to low-ceiling condition. Additionally, overlay-based spatial sharing resulted in reduced temporal demand, task effort, and frustration relative to portal-based sharing. Based on these findings, this paper suggests spatial design guidelines for VR shared spaces that minimize cognitive load and enhance cognitive efficiency in multiplayer VR environments.

Index Terms: Virtual reality, shared space, cognitive load, user

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experience, spatial design.

1 INTRODUCTION

Virtual Reality (VR) has been increasingly applied across domains such as education, training, and remote collaboration due to its ability to overcome spatial constraints. However, the immersive nature of VR can induce high cognitive load depending on how information and spatial elements are designed and presented. Cognitive load refers to the mental resources required by working memory during task execution. Excessive cognitive load often leads to reduced task performance, physical fatigue, and diminished immersion. In particular, in shared VR collaboration spaces where multiple users interact simultaneously, the complexity of visual information and the spatial arrangements can increase users' cognitive load [1]. Despite this, empirical studies examining how physical properties of virtual spaces influence cognitive load remain limited.

This study experimentally analyzes the effects of ceiling height and spatial sharing methods on cognitive load during collaborative tasks in VR shared spaces. By identifying design variables that minimize cognitive load, this research aims to suggest spatial design guidelines to enhance efficiency and user experience in future VR collaborative platforms.

2 METHODS AND MATERIALS

In this study, two virtual rooms were designed as experimental stimuli. Each room employed a distinct interior design to simulate a scenario in which users share a personalized space (Fig. 1a). To minimize potential confounding effects of cognitive load arising from

differences in interior design, both rooms contained the same types and number of objects, and the spatial positions of these objects were kept identical across rooms. Furthermore, to eliminate the influence of environmental factors such as lighting, illuminance, and light direction on cognitive load, identical lighting conditions were applied to both virtual rooms. Each participant of a dyad occupied a different one of the two virtual rooms. When the rooms are shared, they are spatially arranged such that the participants appear to be seated facing each other.

The shared VR environment using these two rooms was designed based on two key experimental variables. The first concerns the spatial sharing methods, and the second concerns the ceiling height of the shared space. We constructed multiple shared-space conditions by combining these variables. To examine the effects of each condition on users' cognitive load during collaboration, participants performed a block-building task in shared spaces.

2.1 Spatial Sharing Methods

Spatial sharing method refers to the approach in which a user's personal space is shared with other users in VR. In this study, we comparatively analyze two commonly adopted approaches [2]: the overlay method and the portal method. The *overlay* method integrates two users' spaces into a single shared environment (Fig. 1b). The *portal* method, on the other hand, uses screen-like digital interfaces—such as windows or screens—to provide visual access to separate virtual spaces (Fig. 1c). In this experiment, the portal was presented in the form of a rectangular window. Its horizontal width was fixed at 1 m, while the vertical length extended from each participant's table to the ceiling. The portal was positioned directly in front of the participant, centered relative to the table.

2.2 Ceiling Height

The height of the ceiling in shared virtual spaces has been identified as an important variable influencing spatial perception and user experience [3, 5]. This study conducts a comparative analysis of three spatial configurations: low-ceiling, high-ceiling, and fully open-ceiling spaces (see Fig. 1b). According to the 'Rules on Housing Construction Standards' in South Korea, the minimum ceiling height for living rooms and bedrooms is specified as 2.2 m. Reflecting current residential conditions in Korea, this study defined the low-ceiling condition as 2.2 m. The high-ceiling condition was set to 3.0 m, considering the high-ceiling height adopted in prior studies [4].

2.3 Collaborative Block-Building Task

This study investigated cognitive load in collaborative settings using a block-building task in which two participants assumed complementary roles. In each task, participants worked in pairs to complete a block-building activity, with one participant assigned the role of instructor and the other the role of executor. The *instructor* had visual access to the target block structure. In contrast, the *executor* could not see the structure and relied solely on the instructor's verbal instructions to assemble the blocks. To minimize role-related participation bias and ensure balanced interaction, participants alternated roles across task trials.

The blocks used in the assembly tasks consisted of four rectangular types: 2x1, 2x2, 2x4, and 4x1 (Fig. 1d). Target block structures for each task were generated using a custom Python-based algorithm. The algorithm randomly selected seven blocks from the four block types, applied random rotations to each, and placed them in a 4x4 grid (Fig. 1d). Using this procedure, a total of 24 distinct target patterns were generated. The difficulty of the patterns was calibrated such that each could be completed within approximately two minutes on average.

Prior to the main experiment, participants completed a set of practice tasks to familiarize themselves with the task rules and

block manipulation procedures. The practice phase consisted of two tasks involving two blocks and two tasks involving seven blocks, presented sequentially. The practice tasks followed the same procedure as the main experiment; however, data collected during this phase were excluded from subsequent analyses.

2.4 Measures

During the experiment, all participants were equipped with a Meta Quest Pro headset, which provided a horizontal field of view of 106° and a vertical field of view of 95.57°. Task completion time and task accuracy were recorded for each trial.

To measure cognitive load under each spatial condition, we used the NASA Task Load Index (NASA-TLX). This questionnaire assesses perceived workload across six dimensions: mental demand, physical demand, temporal demand, performance, effort, and frustration. Participants completed the questionnaire after finishing each spatial condition.

3 RESULTS

A total of 24 participants (12 female; *mean age* = 25.83, *SD* = 3.66) were included in the analysis. A 3 (Ceiling height: High, Low, Open) × 2 (Spatial sharing method: Portal, Overlay) within-subjects design was employed. Depending on whether the assumptions of normality and sphericity were satisfied, either a repeated-measures ANOVA or an aligned rank transform ANOVA was conducted.

The analysis showed a significant main effect of ceiling height across all six NASA-TLX subscales: mental demand, physical demand, temporal demand, performance, effort, and frustration (all *ps* < .01). Post-hoc comparisons indicated that the high-ceiling condition resulted in significantly lower workload ratings than the low-ceiling condition for all six factors: mental demand (*p* < .001), physical demand (*p* = .001), temporal demand (*p* < .001), performance (*p* = .050), effort (*p* = .001), and frustration (*p* = .016). In addition, the open-ceiling condition yielded significantly lower workload ratings than the low-ceiling condition for performance (*p* = .014), effort (*p* = .025), and frustration (*p* = .013). No significant differences were observed between the high- and open-ceiling conditions for these factors. These results indicate that low-ceiling height consistently increased perceived workload, whereas high- or open-ceiling configurations were associated with reduced subjective workload across multiple dimensions.

A significant main effect of spatial sharing method was observed only for temporal demand and frustration (both *ps* < .05). The overlay condition resulted in significantly lower ratings than the portal condition for temporal demand (*p* = .046) and frustration (*p* = .021). No significant main effects of the spatial sharing method were found for other factors. These findings suggest that the choice of spatial sharing method selectively influenced time-related workload and affective responses, while having a limited impact on other workload dimensions.

No significant interaction between ceiling height and spatial sharing method was observed for any of the NASA-TLX factors (all *ps* > .05).

4 DISCUSSION

This study presents initial findings on how ceiling height and spatial sharing methods in VR-based collaborative tasks relate to perceived cognitive load. The results indicate that high- or open-ceiling conditions are associated with lower reported cognitive load compared to low-ceiling conditions. One possible interpretation is that increased vertical openness affords a greater sense of spatial freedom, which may reduce environmental constraints and attentional demands during task execution [4]. To further contextualize these findings, future analyses will examine objective performance measures such as reaction time and task accuracy, which may clarify

how reductions in perceived cognitive load relate to collaborative efficiency and task outcomes.

With respect to spatial sharing methods, the overlay condition was associated with lower temporal demand and frustration than the portal condition. This pattern suggests that overlay-based representations may support more continuous and integrated awareness of collaborators and task context, thereby reducing the cognitive effort required for temporal coordination. However, the extent to which these differences in cognitive load are accompanied by variations in users' experiential qualities remains to be explored. Accordingly, additional analyses incorporating measures of spatial presence and social presence are expected to provide further insight into how different spatial sharing designs shape not only cognitive demands but also users' sense of immersion and social connectedness in collaborative VR environments.

These findings have implications for the design of extended reality (XR) systems that aim to capture, reconstruct, and replay collaborative spatial experiences as forms of spatial memory. Environmental factors such as ceiling height and spatial sharing representations may shape not only users' moment-to-moment cognitive load but also how richly and accurately collaborative episodes are encoded, retained, and later recalled. Lower cognitive load and reduced temporal demand and frustration, as observed in the high-ceiling and overlay conditions, may support deeper attentional engagement with spatial layouts, partner actions, and task-relevant events, potentially leading to more durable spatial memory. Future work could therefore examine how these spatial design parameters affect not only cognitive load but also the subjective vividness and persistence of remembered collaborative scenes, thereby more directly linking collaborative VR design to the formation and retrieval of spatial memories in XR environments.

ACKNOWLEDGMENTS

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (RS-2025-00521923), and by STEAM R&D Project, National Research Foundation (NRF), Korea (RS-2024-00454458).

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